

The Normal Distribution

ANSWER KEY



Properties of the normal distribution

- 1 State three key properties of a normal distribution's shape.

A normal distribution is: (1) symmetric and bell-shaped, centered at its mean; (2) unimodal (a single peak at the mean); (3) its total area under the curve equals 1, and it never touches the horizontal axis (it extends infinitely in both directions, getting closer and closer to zero).

- 2 Explain what the two parameters μ and σ control in a normal distribution's shape.

μ (the mean) controls the location of the center of the distribution (where the peak is). σ (the standard deviation) controls the spread: a larger σ produces a wider, flatter curve, while a smaller σ produces a narrower, taller curve — but both remain symmetric and bell-shaped.

The 68-95-99.7 rule

- 3 Adult male heights are approximately normal with mean 70 inches and standard deviation 3 inches. Using the 68-95-99.7 rule, find the range of heights containing the middle 95% of adult men.

The 95% range spans $\mu \pm 2\sigma = 70 \pm 2(3) = 70 \pm 6$, so from 64 to 76 inches.

- 4 Using the same distribution, find the approximate percentage of adult men taller than 76 inches.

76 inches is $\mu + 2\sigma$. By the 68-95-99.7 rule, 95% of the data lies within 2 standard deviations, leaving 5% split evenly in the two tails: 2.5% above 76 inches and 2.5% below 64 inches.

Standardizing and using the normal distribution

- 5 Using the same height distribution (mean 70, standard deviation 3), find the z-score for a height of 77.5 inches, and use a standard normal table value of $P(Z < 2.5) \approx 0.9938$ to find the proportion of men shorter than 77.5 inches.

$z = \frac{77.5 - 70}{3} = \frac{7.5}{3} = 2.5$. $P(X < 77.5) = P(Z < 2.5) \approx 0.9938$, or about 99.38% of men are shorter than 77.5 inches.

- 6 Find the proportion of men between 65 and 75 inches tall, given $P(Z < 1.67) \approx 0.9525$ and $P(Z < -1.67) \approx 0.0475$.

$z_{65} = \frac{65 - 70}{3} \approx -1.67$. $z_{75} = \frac{75 - 70}{3} \approx 1.67$.

$P(65 < X < 75) = P(Z < 1.67) - P(Z < -1.67) \approx 0.9525 - 0.0475 = 0.905$, or about 90.5%.

Visualizing the standard normal distribution

- 7 The graph shows the standard normal distribution ($\mu = 0$, $\sigma = 1$), with the area to the left of $z = 1$ shaded (though shading itself is not shown, imagine it visually).
- a Using $P(Z < 1) \approx 0.8413$, find the area under the curve to the right of $z = 1$.
 $P(Z > 1) = 1 - P(Z < 1) = 1 - 0.8413 = 0.1587$.
- b Explain, using the curve's symmetry, why $P(Z < -1) = P(Z > 1)$. The standard normal curve is symmetric about $z = 0$. By symmetry, the area to the left of -1 is a mirror image of the area to the right of 1 , so $P(Z < -1)$ and $P(Z > 1)$ must be equal.
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Working backward: finding values from percentiles

- 8 SAT scores are approximately normal with mean 1050 and standard deviation 150. A student scores at the 90th percentile, corresponding to $z \approx 1.28$. Find that student's approximate score.
 $x = \mu + z\sigma = 1050 + 1.28(150) = 1050 + 192 = 1242$.
- 9 Using the same distribution, find the score that separates the bottom 10% of scores, given that $z \approx -1.28$ corresponds to the 10th percentile.
 $x = \mu + z\sigma = 1050 + (-1.28)(150) = 1050 - 192 = 858$.
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Assessing normality

- 10 Explain how a normal probability plot (or normal quantile plot) can be used to assess whether a data set is approximately normal, and what a roughly straight-line pattern indicates.
- A normal probability plot compares the ordered data values to what would be expected if the data were exactly normal. If the plotted points fall close to a straight line, the data is approximately normal. Systematic curvature or bending away from a straight line (such as an S-shape or clear bends at the ends) suggests the data deviates from normality — for example, skewness or heavy tails.
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Applying the normal distribution to real data

- 11 Pregnancy lengths are approximately normal with mean $\mu = 266$ days and standard deviation $\sigma = 16$ days. Find the z-score for a pregnancy lasting 250 days.
 $z = \frac{250 - 266}{16} = \frac{-16}{16} = -1$.
- 12 Using the same pregnancy-length distribution and the 68-95-99.7 rule, find the approximate percentage of pregnancies lasting between 234 and 298 days.
 $234 = 266 - 32 = \mu - 2\sigma$ and $298 = 266 + 32 = \mu + 2\sigma$, so this range spans $\mu \pm 2\sigma$. By the 68-95-99.7 rule, approximately 95% of pregnancies fall in this range.
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Combining two independent normal random variables

- 13** A vending machine dispenses a soda whose volume is normally distributed with mean 355 mL and standard deviation 4 mL, independently of a snack bag whose weight is normally distributed with mean 50 g and standard deviation 2 g. If T represents the combined "total" formed by adding the soda volume (in mL) and snack weight (in g) as a single number, find the mean of T . (This is a simplified illustrative combination, not a real physical unit.)

For independent normal random variables, means add directly: $\mu_T = 355 + 50 = 405$.

- 14** Battery lifespans are normally distributed with mean 500 hours and standard deviation 40 hours. Find the z-score for a battery lasting 460 hours, and describe its relative position.

$z = \frac{460 - 500}{40} = \frac{-40}{40} = -1$. This battery's lifespan is exactly one standard deviation below the mean.

- 15** A company manufactures resistors with resistance values approximately normal, mean 100 ohms and standard deviation 5 ohms. A resistor is considered defective if its resistance is below 90 ohms or above 110 ohms. Using the 68-95-99.7 rule, estimate the percentage of resistors that are defective, and explain your reasoning step by step.

$90 = 100 - 2(5) = \mu - 2\sigma$ and $110 = 100 + 2(5) = \mu + 2\sigma$, so the acceptable range spans $\mu \pm 2\sigma$. By the 68-95-99.7 rule, approximately 95% of resistors fall within this range (not defective), leaving approximately 5% defective (split as roughly 2.5% below 90 ohms and 2.5% above 110 ohms).

- 16** Two students are comparing their standardized test scores from different tests. Student X scored 78 on a test with mean 70 and standard deviation 8. Student Y scored 165 on a different test with mean 150 and standard deviation 12. Using z-scores, determine who performed relatively better compared to their own test's distribution of test-takers, showing full reasoning.

Student X: $z = \frac{78 - 70}{8} = \frac{8}{8} = 1$. Student Y: $z = \frac{165 - 150}{12} = \frac{15}{12} = 1.25$. Since $1.25 > 1$, Student Y performed relatively better compared to their own test's distribution of test-takers, even though the raw score gap looks different for each test.